

1.22

$$C_6^3 \cdot C_3^3 \cdot C_6^4 \cdot C_1^1 = 20 \cdot 1 \cdot 15 \cdot 1 = \underline{300}$$

2.22

$$\frac{1}{5} \cdot \frac{1}{4} = \frac{1}{20} = \underline{0,05}$$

Примечание: В ответе написано  $0,0167 \approx \frac{1}{60}$  -

вероятность 3 букв подряд. В задании просят 2 буквы.

3.22

$$a) \frac{7}{20} \cdot \frac{4}{30} = \frac{7}{150} \approx \underline{0,04667}$$

$$b) \frac{7}{20} \cdot \frac{26}{30} + \frac{13}{20} \cdot \frac{4}{30} = \frac{224}{600} = \underline{0,373}$$

$$b) \frac{13}{20} \cdot \frac{26}{30} = \frac{338}{600} \approx \underline{0,5633}$$

4.22

$$a) \frac{3}{8} \cdot \frac{6}{10} + \frac{5}{8} \cdot \frac{5}{10} = \underline{0,5375}$$

$$b) \text{ купил в А } \frac{3}{8} \cdot \frac{6}{10} = 0,225$$

$$\text{купил в В } \frac{5}{8} \cdot \frac{5}{10} = 0,3125$$

$$\frac{0,225}{0,5375} \approx 0,4186$$

$$\frac{0,3125}{0,5375} \approx 0,5814$$

в кассе возврата В



5.22

Барининев Т.А. Р3270 А

$$a) 1 - (1 - 0,8)^8 \approx \underline{0,999997}$$

$$b) \frac{8!}{2! \cdot 6!} \cdot 0,8^2 \cdot (1 - 0,8)^6 \approx \underline{0,001146}$$

$$b) \frac{8!}{3! \cdot 5!} \cdot 0,8^3 \cdot (1 - 0,8)^5 \approx \underline{0,00917}$$

6.22

$$P\{X=k\} \approx \frac{1}{\sqrt{npq}} \cdot \varphi(x)$$

$$x = \frac{k - np}{\sqrt{npq}}$$

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

$$np = 1000 \cdot 0,5 = 500$$

$$npq = 1000 \cdot 0,5 \cdot 0,5 = 250$$

$$x = \frac{500 - 500}{\sqrt{250}} = 0$$

$$\varphi(0) = \frac{1}{\sqrt{2\pi}} \approx 0,39894$$

$$P\{X=500\} \approx \frac{1}{5\sqrt{10}} \cdot \frac{1}{\sqrt{2\pi}} = \frac{1}{5\sqrt{20\pi}} \approx \underline{0,0252}$$

УДЗ 18-2

7.22

$$M(X) = 3 \cdot 0,3 = 0,9$$

$$D(X) = 3 \cdot 0,3 \cdot (1 - 0,3) = 0,63$$



2.22

Пример 2.2. P3210

$$f(x) = \begin{cases} 0 & x < 2 \\ \frac{2}{3}(x-1) & 2 \leq x \leq 3 \\ 0 & x > 3 \end{cases}$$

$$M(x) = \int_2^3 x \cdot \frac{2}{3}(x-1) dx = \frac{2}{3} \int_2^3 (x^2 - x) dx = \frac{2}{3} \left[ \frac{x^3}{3} - \frac{x^2}{2} \right]_2^3 =$$

$$= \frac{2}{3} \left( \frac{27}{3} - \frac{9}{2} - \frac{8}{3} + \frac{4}{2} \right) = \frac{2}{3} \cdot \frac{23}{6} = \frac{23}{9} \approx \underline{2.56}$$

$$M(x^2) = \int_2^3 x^2 \cdot \frac{2}{3}(x-1) dx = \frac{2}{3} \int_2^3 (x^3 - x^2) dx = \frac{2}{3} \left[ \frac{x^4}{4} - \frac{x^3}{3} \right]_2^3 =$$

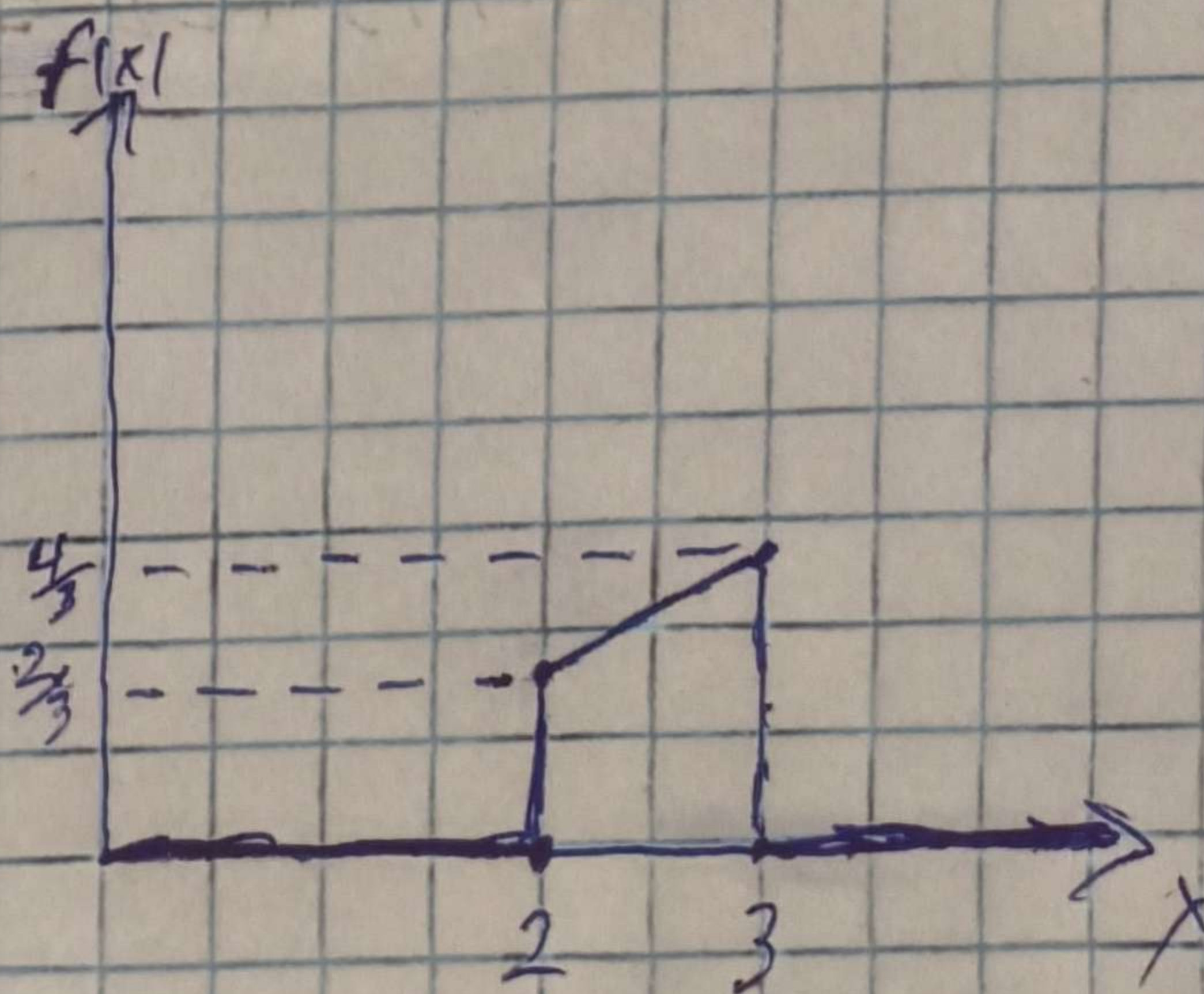
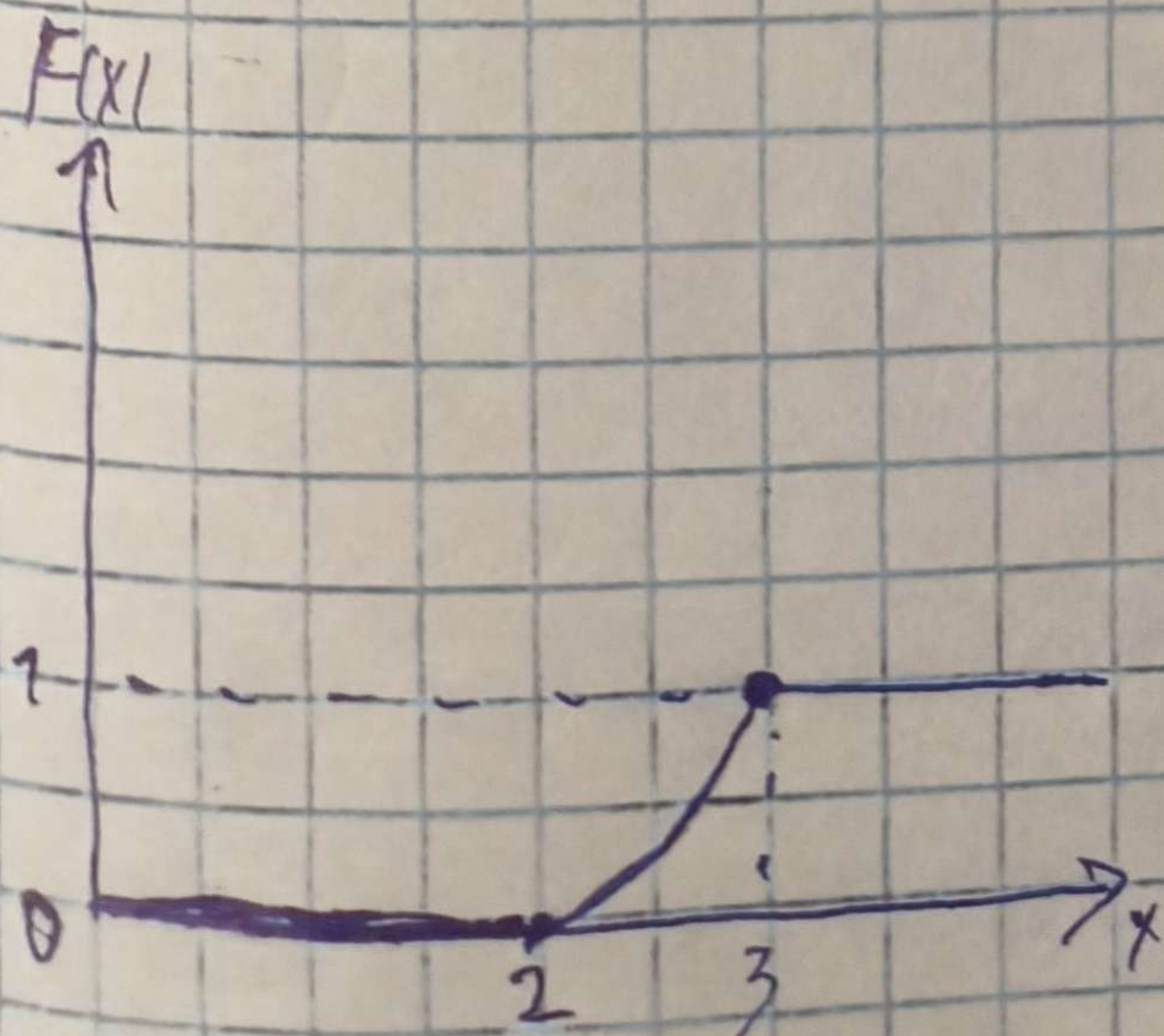
$$= \frac{2}{3} \left( \frac{81}{4} - \frac{27}{3} - \frac{16}{4} + \frac{8}{3} \right) = \frac{2}{3} \cdot \frac{119}{12} = \frac{119}{18} \approx \underline{6.6111}$$

$$D(x) = 6.6111 - 6.5536 = 0.0575 \approx \underline{0.06}$$

$$F(2.5) = \frac{1}{3} [(2.5)^2 - 2 \cdot 2.5] = \frac{1.25}{3} = \underline{0.4167}$$

$$F(2.2) = \frac{1}{3} [(2.2)^2 - 2 \cdot 2.2] = \frac{0.44}{3} = \underline{0.1467}$$

$$P(2.2 \leq x \leq 2.5) = 0.4167 - 0.1467 = \underline{0.27}$$





3.22

Бармичев Т.А. Р3210

$$Z_1 = \frac{1-1,5}{0,04} = -12,5$$

$$Z_2 = \frac{2-1,5}{0,04} = 12,5$$

$$P(1 \leq x \leq 2) = P(-12,5 \leq Z \leq 12,5) = \Phi(12,5) - \Phi(-12,5) = 1 - 0 = 1$$

$$\Phi(12,5) \approx 1 - \frac{0,4 \cdot 10^{-35}}{12,5} \approx 1$$

$$\Phi(-12,5) \approx 1 - 1 = 0$$

4.22

$$M(X) = 3000 \cdot 0,75 = 2250$$

$$D(X) = 3000 \cdot 0,75 \cdot (1 - 0,75) = 562,5$$

$$|2250 - 2290| = |2250 - 2310| = 60$$

$$P(|X - 2250| \leq 60) \approx 1 - \frac{562,5}{60^2} = \underline{0,84375}$$